

Site “Cedar” Bai

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EDUCATION

- **Purdue University** Jan. 2021 – Dec. 2025 (Exp.)
Ph.D. in Computer Science (Machine Learning / Optimization) Grade: 3.9/4.0
M.S. in Statistics & Computer Science (along the way) M.S. conferred May. 2024
- **Xi'an Jiaotong University (XJTU)** Sep. 2016 – Jun. 2020
B.S. in Computer Science (Qian Xuesen Honors College) Grade: 85.7/100
Selected Awards: “Siyuan” Scholarship (twice); National 2nd Prize in Mathematical Contest in Modeling of China (CUMCM, top 3%)

PROFESSIONAL TECHNIQUES

Machine learning (ML): ML optimizer design/analysis, deep reinforcement learning, federated learning, recommendation, unlearning
Programming: Python, MATLAB (**Proficient**); Shell script, C/C++, R, Java, SQL, HTML;
Packages / Tools: PyTorch, NumPy, Matplotlib, Linux Command, Git (**Proficient**); scikit-learn, TensorFlow, Pandas, Slurm.

INDUSTRY EXPERIENCE

- **TikTok, Machine Learning Engineer Intern** May. 2024 – Aug. 2024
Worked on **ranking models in ads recommendation systems** Manager: *Du Zhang, Dr. Yan Yan*
 - Delivered two launches that improved advertiser value by 1.63% and 5.94%, a **\$X0,000 increase in daily profits**
 - Applied neural architecture search to select the best features and embedding sizes, reducing the model size while improving accuracy
 - Adapted Deep Cross Network as the feature crossing module for improved conversion rate (CVR) prediction on sequential data

SELECTED PROJECTS

- ▷ Department of Computer Science, Purdue Advisor: *Prof. Brian Bullins*
 - **Deep Learning Optimization | Faster Acceleration for Steepest Descent**
 - Proposed accelerated steepest descent for ℓ_p -smooth functions based on the interpolation of steepest descent and mirror descent
 - Derived convergence rates for convex and (stochastic) non-convex settings, and implemented the optimizer with PyTorch
 - Achieved faster convergence and higher precision than Adam and Lion on **ImageNet and pre-training large language models**
 - **Optimization & Federated Learning | Federated Composite Saddle Point Optimization**
 - Proposed the first distributed optimization algorithm for min-max problems with non-smooth structure-inducing regularization
 - Derived convergence rate, wrote the optimizer with PyTorch, and conducted distributed learning experiments with FedLab
 - Applied the optimizer to the **universal adversarial training** on l_1 -regularized logistic regression with real-world data
 - Achieved **faster convergence and better structure** (e.g. sparse solution) than (projected) gradient-descent-ascent
- ▷ Institute of Artificial Intelligence and Robotics, XJTU Advisor: *Dr. Hanbo Zhang, Prof. Xuguang Lan*
 - **Deep Reinforcement Learning | Hindsight Trust Region Policy Optimization**
 - Proposed a deep reinforcement learning algorithm that improves **sample efficiency** in environments with **sparse reward**;
 - Applied goal-conditioned policy with (**weighted**) **importance sampling** to learn with both succeeded samples and failed ones.
 - Proposed a Quadratic KL constraint with **math-provable variance reduction** to stabilize training;
 - Implemented experiments with PyTorch on games with image input and robot motion planning simulation in OpenAI Gym.

PUBLICATION

- [1] [Preprint] **S. Bai***, Y. Zheng*, B. Bullins, R. Yeh. Model Immunization from a Condition Number Perspective. 2025.
- [2] [Preprint] **S. Bai***, X. Luo*, B. Li*, P. Drineas, R. Zhang, B. Bullins. Stochastic Steepest Descent with Acceleration for ℓ_p -Smooth Non-convex Optimization. 2024. [\[Abstract\]](#)
- [3] [Preprint] **S. Bai**, B. Bullins. Faster Acceleration for Steepest Descent. 2024. [\[PDF\]](#)
- [4] [ICLR 2025, **Oral (Top 1.7%)**] **S. Bai**, B. Bullins. Tight Lower Bounds under Asymmetric High-Order Hölder Smoothness and Uniform Convexity. [\[PDF\]](#)
- [5] [ICLR 2024] **S. Bai**, B. Bullins. Local Composite Saddle Point Optimization. [\[PDF\]](#)
- [6] [TMLR 2024] **S. Bai**, C. Ke, J. Honorio. On the Dual Problem of Convexified Convolutional Neural Networks. [\[PDF\]](#)
- [7] [IJCAI 2021] H. Zhang, **S. Bai**, X. Lan, D. Hsu, N. Zheng. Hindsight Trust Region Policy Optimization. [\[PDF\]](#)
- [8] [IROS 2019] H. Zhang, X. Lan, **S. Bai**, X. Zhou, Z. Tian, N. Zheng. ROI-based Robotic Grasp Detection for Object Overlapping Scenes. [\[PDF\]](#)
- [9] [IROS 2019] H. Zhang, X. Lan, **S. Bai**, L. Wan, C. Yang, N. Zheng. A Multi-task Convolutional Neural Network for Autonomous Robotic Grasping in Object Stacking Scenes. [\[PDF\]](#)

SELECTED COURSES

Machine Learning (ML): Statistical ML, ML Theory, Computer Vision, Data Mining, Artificial Intelligence, etc.;

Math & Statistics: Optimization, Probability, Statistical Inference, Computational Statistics, Multivariate Analysis, Quantitative Finance;

Computer Science: Data Structures, Algorithms, Database, Machine Structure, Operating Systems, Networks, etc.

TEACHING & ACADEMIC SERVICE

Purdue CS571 Artificial Intelligence	TA (Fall 23)
Purdue CS182 Foundations of Computer Science (Discrete Mathematics)	TA (Spring 23)
Purdue CS251 Data Structures and Algorithms	TA (Fall 21, Spring 22, Fall 22)

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| • ICLR 2025, 2024 | Reviewer |
| • NeurIPS 2024, 2023 | Reviewer |
| • ICML 2025, 2024 | Reviewer |
| • AISTATS 2025, 2023 | Reviewer |
| • AAAI 2025 | Reviewer |
| • TMLR | Reviewer |